# Chapter 8

# Biostatistics and Data Analysis

# Contents

- Hypothesis testing problems: Constructing Hypothesis
- Concepts of Hypothesis Testing: Null and Alternative Hypothesis

# Introduction

## Hypothesis Testing

## Definition

- **Hypothesis Testing** is a statistical method used to make decisions or draw conclusions about a **population** based on information from a **sample**.

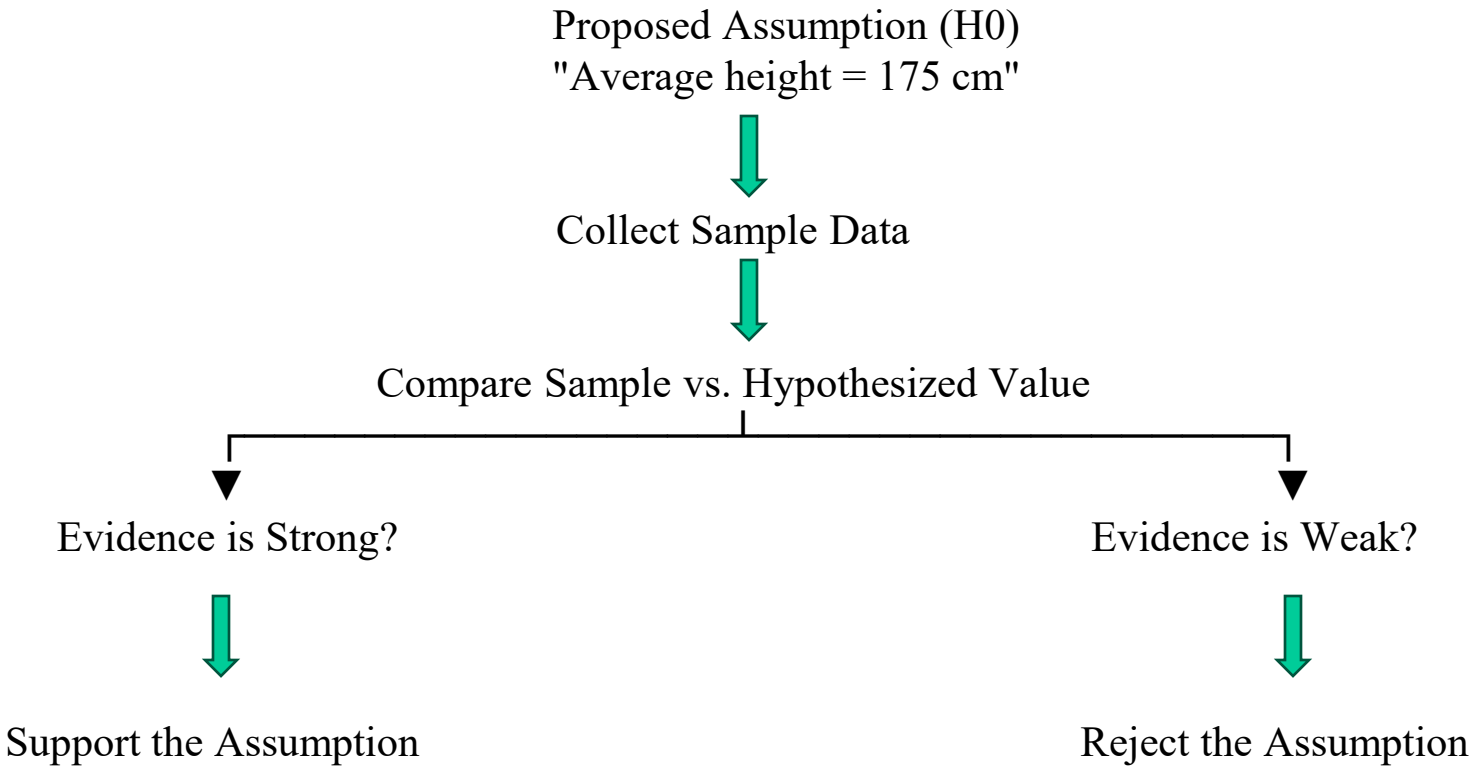
# Introduction

## Hypothesis Testing

- **Goal:**
  - ✓ To evaluate a proposed assumption or belief about a population parameter using sample data.
- **Method:**
  - ✓ We compare the information from a sample to the hypothesized population value.
  - ✓ If the sample does not provide strong enough evidence, we may reject the proposed assumption.
- **Example:**
  - ✓ Suppose we want to test whether the average height of all adults is 175 cm.
  - ✓ By collecting random sample data, we check whether the evidence supports or contradicts this claim.

# Introduction

## Hypothesis Testing



# Hypothesis Testing

**Hypothesis Testing Steps**

1. State Hypotheses
2. Choose a Test Statistic
3. Calculate the Test Statistic
4. Determine the P-value

# Hypothesis Testing

## 1. State Hypotheses

### ➢ Purpose of Hypothesis

✓ The purpose of stating hypotheses is to convert a

**Research Question** into **Testable Statements.**

### ➢ Types of Hypothesis

#### i. Null Hypothesis ($H_0$)

✓ Represents the claim of no effect or no difference in the population.

✓ Symbolically: $H_0$ usually includes "=" (equal, no difference, no change).

#### ii. Alternative Hypothesis ($H_a$)

✓ Represents the **claim that $H_0$ is false**.

✓ Also called the **research hypothesis**.

✓ Indicates the presence of difference, or change.

# Hypothesis Testing

## 1. State Hypotheses

**Types of Alternative Hypotheses:**

- ✓ One sided alternative hypothesis
  - ❖ Tests for a change in **one specific direction** (increase or decrease).
  - ❖ It has a specific direction (>, <)
  - ❖ Uses symbols: > or <
- ✓ Two sided alternative hypothesis
  - ❖ Tests for **any change**, either increase or decrease.
  - ❖ But does not specify direction (≠)
  - ❖ Uses symbol: ≠

# Hypothesis Testing

## 1. State Hypotheses

**Illustrative Example: Body Weight in Dawadmi**

**Background:**

In the 1980s, 20–29-year-old men in Dawadmi had:

- ✓ Mean body weight: μ = 170 pounds (77 kg)
- ✓ Standard deviation: σ = 40 pounds

- **Research Question:**

  Has the mean body weight of 20–29-year-old men in Dawadmi **changed** since the 1980s?

- **Null Hypothesis ($H_0$):**

$$H_0: \mu = 170$$

Meaning: There is **no difference**; the current population mean body weight is the same (170 pounds) as in the 1980s.

# Hypothesis Testing

## 1. State Hypotheses

- **Alternative Hypothesis ($H_a$):**
  - ✓ **One-sided (Directional) Alternative Hypothesis:**

    Test for **increase**: $H_a: \mu > 170$

    Test for **decrease**: $H_a: \mu < 170$

  - ✓ **Two-sided (Non-directional) Alternative Hypothesis:**

    Test for **any change** (increase or decrease): $H_a: \mu \neq 170$

Meaning: The population mean has **changed**, but the direction (increase or decrease) is **not specified**.

# Hypothesis Testing

## 2. Choose the Test Statistic

### Types of Test Statistics

#### 1. z-score (z-test)

- **Use:** Comparing means when **sample size (n) > 30** or population standard deviation (σ) is known.

#### 2. t-statistic (t-test)

- **Use:** Comparing means when population standard deviation is unknown, **irrespective of sample size**.
- More flexible than z-test, especially for smaller samples.

# Hypothesis Testing

## 2. Choose the Test Statistic

## Types of Test Statistics

**Common t-tests**

a. One-sample t-test

- **Purpose:** Compare the mean of a single sample to a **fixed value**.
- **Example:** Testing whether the mean hemoglobin (Hb) level in a sample is equal to 13.5 g/dL.

b. Two independent samples t-test (Independent t-test)

- **Purpose:** Compare means between **two independent groups**.
- **Example:** Comparing Hb levels between males and females.

c. Dependent samples t-test (Paired t-test)

- **Purpose:** Compare means between **two related/dependent groups** (same subjects measured twice).
- **Example:** Comparing Hb levels **before and after treatment** (e.g., after administering Maltofer).

# Hypothesis Testing

## 2. Choose the Test Statistic

## Types of Test Statistics

### 3. ANOVA (Analysis of Variance)

- **Purpose:** Compare **means of three or more independent groups**.
- **Example:** Comparing Hb levels among samples from **Riyadh, Dawadmi, and Shaqra**.
- **Key point:** If ANOVA is significant, it indicates at least one group mean differs, but post-hoc tests are needed to find which.

### 4. Chi-squared ($\chi^2$) statistic

- **Purpose:** Determine if there's a **significant association between two categorical variables**.
- **Example:** Investigating whether taking **cyclophosphamide** affects **5-year mortality rates** (Yes/No).
- **Key point:** Works with counts/frequencies, not continuous data.

# Hypothesis Testing

## 2. Choose the Test Statistic

## Types of Test Statistics

### 5. Pearson correlation coefficient (r)

- **Purpose:** Measure the **strength and direction of the association between two continuous variables**.
- **Example:** Investigating whether **Hb levels change with age**.
- **Key point:** Values range from **-1 to +1**;
  - ✓ Positive values indicate direct association
  - ✓ Negative values indicate inverse association
  - ✓ 0 indicates no linear association.

# Hypothesis Testing

## 2. Choose the Test Statistic

### Types of Test Statistics

| Test Statistic | Purpose | Use Case / When to Use | Example |
|---|---|---|---|
| **z-score (z-test)** | Compare sample mean to population mean | Sample size **n > 30**, population SD known | Test if average weight of a population differs from known value |
| **t-statistic (t-test)** | Compare means when population SD unknown | Any sample size, SD unknown | Test if average Hb differs from 13.5 g/dL |
| **One-sample t-test** | Compare **one sample mean** to a fixed value | Single group comparison | Hb level of patients vs 13.5 g/dL |
| **Two independent samples t-test** | Compare means of **two independent groups** | Two separate groups | Compare Hb between males and females |
| **Paired t-test (dependent samples)** | Compare means of **two related groups** | Same subjects measured twice | Compare Hb **before and after treatment** |
| **ANOVA** | Compare means of **three or more independent groups** | Multiple independent groups | Compare Hb from Riyadh, Dawadmi & Shaqra |
| **Chi-squared ($\chi^2$)** | Test association between **two categorical variables** | Categorical data | Check if cyclophosphamide affects 5-year mortality |
| **Pearson correlation (r)** | Measure **association between two continuous variables** | Continuous variables | Investigate if Hb levels change with age |

# Hypothesis Testing

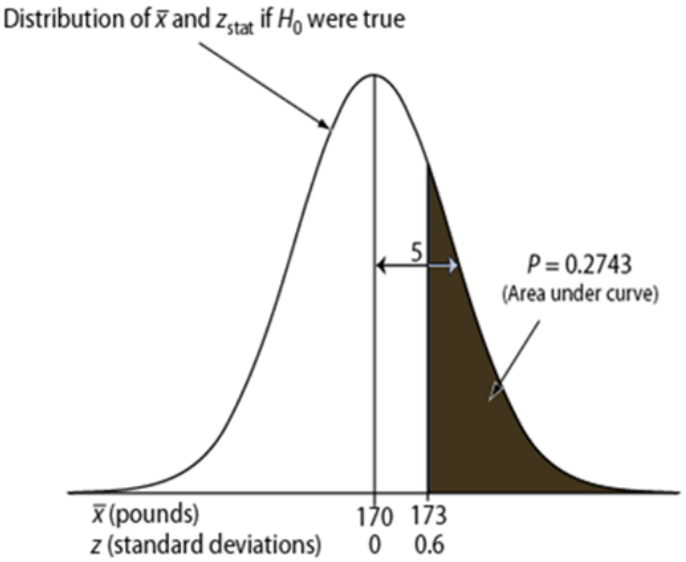
## 3. Calculate the Test Statistic

One-Sample Z-Test (σ known)

**Given:**

- Null Hypothesis ($H_0$): $\mu = \mu 0 = 170$
- Alternative Hypothesis (Ha): $\mu \neq 170$ (two-tailed)
- Population Standard Deviation (σ): 40
- Sample Size (n): 64

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{64}} = 5$$




**Test Statistic:**

$$z = \frac{\bar{x} - \mu_0}{SE}$$

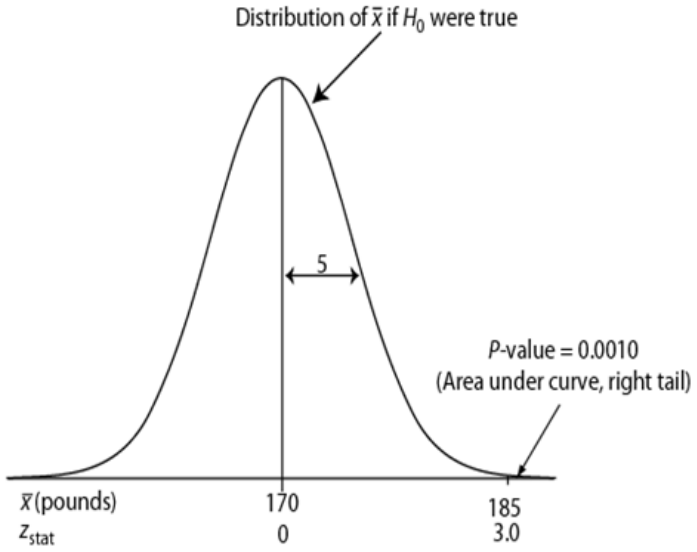
**Calculation:**

- If we found a sample mean ($\bar{x}$) = 173, then

$$z = \frac{\bar{x} - \mu_0}{SE} = \frac{173 - 170}{5} = 0.60$$

- If we found a sample mean ($\bar{x}$) = 185, then

$$z = \frac{\bar{x} - \mu_0}{SE} = \frac{185 - 170}{5} = 3.00$$



# Hypothesis Testing

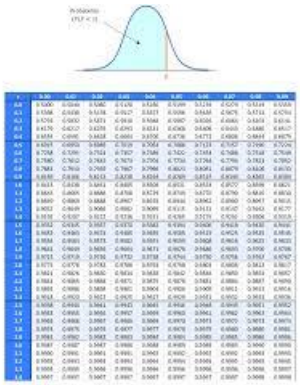
## 4. Determine the P-value

- The P-value represents the probability of observing a test statistic (Z-score) at least as extreme as the one observed, assuming the null hypothesis (H0) is true.

**Breakdown**

1. Probability: Chance or likelihood
2. Observing: Seeing or obtaining
3. Test statistic: Z-score (standardized value)
4. At least as extreme: Equal to or more extreme than
5. Assuming $H_0$ is true: Considering the null hypothesis as correct

# Hypothesis Testing



## 4. Determine the P-value

**Two-tailed test:** $P-value = 2 \times P(Z \geq |\ z\ |)$

- **For $z = 0.6$:**

Using a standard normal distribution table (Z-table) or a calculator:

$$P-value = 2 \times P(Z > |\ 0.6\ |)$$

$$\approx 2 \times 0.2743$$

$$\approx 0.5486$$

✓ **Interpretation:** Since $P > 0.10$ $(0.5486 > 0.10)$, we **fail to reject** $H_0$.

✓ **Conclusion:** The sample mean of **173** is **not significantly different** from 170.

**OR** There is **no evidence** that the mean is different from 170.

**P-value ≥ 0.10:**

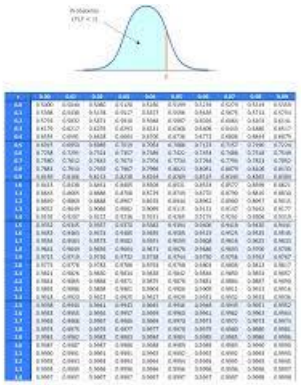
➤ **No evidence** against $H_0$

➤ Result is **not significant**



Note: The result is not significant because we do not have enough evidence to reject $H_0$.

# Hypothesis Testing



## 4. Determine the P-value

**Two-tailed test:** $P-value = 2 \times P(Z \geq |\ z\ |)$

- **For $z = 3$:**

Using a standard normal distribution table (Z-table) or a calculator:

$$P-value = 2 \times P(Z > |\ 3\ |)$$

$$\approx 2 \times 0.00135$$

$$\approx 0.0027$$

✓ **Interpretation:** Since $P < 0.01$ $(0.0027 < 0.01)$, we **can reject** $H_0$.

✓ **Conclusion:** The sample mean of **185** is **significantly different** from 170.

So, we have **strong evidence** that the true mean is **not 170**.

**P-value < 0.01:**

➤**Strong evidence** against $H_0$

➤Result is **highly significant**



Note: The result is highly significant because we do have enough evidence to reject $H_0$.

# Interpretation Guidelines for P-values

**P-value < 0.01:**

➤**Strong evidence** against $H_0$

➤Result is **highly significant**

**0.01 ≤ P-value < 0.05:**

➤ **Moderate evidence** against $H_0$

➤Result is **significant**

**0.05 ≤ P-value < 0.10:**

➤ **Weak evidence** against $H_0$

➤Result is **marginally significant**

**P-value ≥ 0.10:**

➤ **No evidence** against $H_0$

➤Result is **not significant**

# Interpretation Guidelines for P-values

| | |
|---|---|
| P-value $< 0.01$ | ➤ Strong evidence against $H_0$<br>➤ Result is highly significant |
| $0.01 \leq$ P-value $< 0.05$ | ➤ Moderate evidence against $H_0$<br>➤ Result is significant |
| $0.05 \leq$ P-value $< 0.10$ | ➤ Weak evidence against $H_0$<br>➤ Result is marginally significant |
| P-value $\geq 0.10$ | ➤ No evidence against $H_0$<br>➤ Result is not significant |


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